

UNIVERSITY OF CALIFORNIA, SAN DIEGO

CONSENT TO PARTICIPATE IN RESEARCH

1. Study Title and Number

Title: Evaluating the Impact of AI-Enhanced Tools on Creative Design Processes
Study # 810643

2. Principal Investigator

Steven Dow, Professor, Department of Cognitive Science

3. Principal Investigator Phone Number, Research Team Number, and Emergency Contact Number

Dr. Steven Dow: (858) 534-7141, spdow@ucsd.edu

4. Study Overview

This research will evaluate how AI-enhanced tools can impact creative design processes, potentially leading to more efficient and innovative outcomes in various design tasks.

(AI here refers to machine intelligence that mimics human decision-making and enables computer systems to help analyze data of different modalities, such as text, image, video, audio, etc.)

We invite you to participate in a research study because you are interested in creative design activities related to product design, graphic design, or similar fields. We hope your experience and insights will help us understand the effectiveness of AI-enhanced tools in supporting and improving creative design processes.

We recruit participants who are adults over 18 with English proficiency to participate in this study. There are no exclusion criteria other than being able to speak English fluently. We are looking for a wide range of genders, ages, cultural backgrounds, and experience with design, large language models (LLM), machine learning, task domain knowledge, etc.

This form explains the research so that you may make an informed decision about participating.

- Research is voluntary—whether or not to participate is your decision. You can discuss your decision with others (such as family or friends).
- You can say yes, but change your mind later.
- If you say no, we will not hold your decision against you.
- You will be given a digital copy of this consent form.

This research study aims to assess the impact of AI-enhanced tools on the creative design process, focusing on efficiency, accuracy, and user satisfaction.

3 main categories of risks are:

Physical Risks:

- Eye strain or fatigue from prolonged use of computer screens during the design tasks. This common issue can typically be mitigated by taking regular breaks.
- Potential for repetitive strain injuries if the participant has to use a mouse or keyboard extensively.

Psychological Risks:

- Frustration or anxiety if the AI-enhanced tool does not function as expected or if the participant feels pressure to complete the design task within a set timeframe. This is generally mild and temporary.

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- Stress related to performance: Participants might experience stress if they feel their performance is being evaluated, although this study is designed to minimize such pressure.

Risks of Loss of Confidential Information:

- There is also a risk that your information could be released to an unauthorized party. To minimize this risk, we will keep a link between the anonymization code, anonymized data, and your identity in different locations.

You will be required to visit our facility in person or virtually on Zoom once during the study, with each visit lasting approximately 1 hour. The pre-experiment steps include a 5-minute screening survey to gather demographic information and backgrounds in design, Generative AI, and creativity support tools. Qualified candidates will be assigned a convenient one-hour experiment time slot. The 1-hour experiment consists of giving a consent form (5 mins), receiving instructions (5 mins), reading design materials (10 mins), undergoing system training (5 mins), interacting with the assigned tool for the design task (20 mins), and participating in a short survey and interview (10 mins). The total session aims to be completed within an hour.

We cannot promise you or others any benefit from participating in this research. However, possible benefits include enhanced skills in using AI tools for design tasks and gaining insights into new design processes. Additionally, your participation will contribute to developing more effective AI-enhanced design tools, benefiting the broader design community and society.

More detailed information about this research study is provided below.

5. Whom can I talk to if I have questions?

If you have questions or concerns during your participation in the study, or if you think the research has hurt you, contact the research team at 858-226-2323 (Sirui). You should not agree to participate in this study until the research team has answered any questions about the study, including information in this form.

If before or during your participation in the study, you have questions about your rights as a research participant, or you want to talk to someone outside the research team, please contact:

- UC San Diego Office of IRB Administration at 858-246-4777 or irb@ucsd.edu

6. How many people will take part?

We plan to study 200 people here. The research will include UCSD students we recruited from Sona.

7. What happens if I take part in the research?

Here is what will happen to you if you agree to be in this study:

You will be part of a study testing a design system enhanced with artificial intelligence (AI). This system is designed to improve the process and outcome of a design task.

As you read this form, ask questions if something is not clear.

Visit Schedule, Duration, and Location:

- Visit Schedule: You must attend an in-person location or the Zoom session once during the study at the required time. (Only one of the two options)
- Duration: Each visit will last approximately 1 hour.

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- Location: All visits will occur at the Design and Innovation building's 3rd-floor meeting room or virtually on Zoom.

Procedures:

- Pre-Experiment Steps
 - Screening Survey: This survey collects demographic information and participants' backgrounds in design, Generative AI, and creativity support tools. This will take approximately 5 minutes.
 - Timeslot allocation: A qualified candidate (who can speak English satisfactorily) will be assigned a one-hour experiment time slot that works for them.
- The 1-hour experiment process is composed of:
 1. Give the consent form (5 mins)
 2. Instruction (5 mins): Explain the task of designing an everyday product by exploring multiple possibilities and ultimately selecting a favorite design.
 3. Read documents (10 mins): Participants will read through the provided design materials (design requirements, design notes, user persona, and inspirational images). They can read aloud or just skim through the documents in whatever way they feel most comfortable.
 4. System Training (5 mins): Participants watch an introduction video about the UI condition they will be exposed to (one of the A/B/C conditions mentioned above). All these three UI conditions will have varying levels of AI technology packaged in different user interfaces.
 5. Interaction Setup (20 mins): Participants interact with the assigned tool.
 - a. Explore different designs.
 - b. Pick one or a few of their ideal design from all the images they added to their favorite folder.
 - c. Describe the reasoning for the ideal chair choice in text format.
 6. Survey & Interview (10 mins): Participants will complete a self-evaluated survey and can optionally choose to answer some general user experience questions about the tool if they are open to it.

(In a total of 55 minutes, we aim to finish the session within an hour.)

Randomization:

Randomization means that you are randomly assigned to a group—like flipping a coin. Neither you nor the researchers choose which group you will be in. You have a $\frac{1}{3}$ (~33%) chance of being placed in a specific group.

- Group A: Using the AI-enhanced generative design system with full functionalities.
- Group B: Using the baseline system 1 - AI-enhanced generative design system with only contextual information extraction and image generation capability.
- Group C: Using the baseline system 2 - only image generation capability.

Confidentiality:

Your personally identifiable information, such as your name, will be separated from the research data collected and stored securely. Only authorized personnel will have access to it. We will destroy the link between those two when the research is concluded.

The data we store in our data repository includes demographic information, the research subject identification number, images you generated, the text prompt and tags you used to generate the image, the modifications to the text prompt and tags, the time it takes to finish each interaction, and your post-experiment survey responses.

8. What are the risks and possible discomforts?

Participation in this study may involve risks or discomforts.

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Physical Risks:

- Eye strain or fatigue from prolonged use of computer screens during the design tasks. This common issue can typically be mitigated by taking regular breaks.
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Psychological Risks:

- Frustration or anxiety if the AI-enhanced tool does not function as expected or if the participant feels pressure to complete the design task within a set timeframe. This is generally mild and temporary.
- Stress related to performance: Participants might experience stress if they feel their performance is being evaluated, although this study is designed to minimize such pressure.

Risks of Loss of Confidential Information:

- There is also a risk that your information could be released to an unauthorized party. To minimize this risk, we will keep a link between the anonymization code, anonymized data, and your identity in different locations.

9. How will information about me be protected?

While we cannot guarantee complete confidentiality, we will limit access to information about you. Only people who need to review your information, documents, or generated results will have access. These people might include:

- Members of the research team and other staff or representatives of UCSD whose work is related to the research or to protecting your rights and safety.

To further protect your confidentiality, the study information will be labeled with a code instead of your name or other information that can easily identify you. The record linking your identifying information (name, address, etc.) and the code will be kept separate from the rest of the study information.

The results of this study may be published once it is completed. However, we will keep your name and other identifying information confidential. We expect this study will be completed in _3_ months. This is only an estimate, and the time required to complete the study may be longer or shorter depending on several factors.

10. Will I need to pay to participate in the research?

There will be no cost to you to participate in this study.

11. What if I agree to participate but change my mind later?

You can stop participating anytime for any reason, and it will not be held against you. No matter what you decide, there will be no penalty to you. You will not lose anything or any legal rights.

If you stop early, please contact us immediately. We will ask you to give us brief feedback on the study and explain why you would like to opt-out.

12. What will happen to the information collected from me?

The data we collect with your identifiable information (for example, your name or date of birth) as a part of this study may be used to answer other research questions or shared with other investigators for other research. If so, we will remove all identifiable information before use or sharing. Once identifiers have been removed, we will not ask for your consent to use or share your data in other research. In addition, data that have been de-identified will be uploaded to a cloud storage system shared among members of the research team for other researchers to access and use.

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The data that will be uploaded to a long-term cloud storage system include demographic information, the research subject identification number, images you generated, the text prompt and tags you used to generate the image, the modifications to the text prompt and tags, the time it takes to finish each interaction, and your post-experiment survey responses.

13. Will I be compensated for participating in the research?

We recruit participants in this research from SONA through UCSD; we will not provide monetary compensation.

This study can be conducted remotely via Zoom or in person at the Design and Innovation Building's 3rd-floor meeting room. The Design and Innovation Building is located at 9510 Innovation Ln, La Jolla, CA 92093. The compensation will be 0.5 course credits for the remote session and 2 course credits for the in-person session. You can only choose one of the options.

We will not reimburse any out-of-pocket expenses related to your participation, such as travel costs.